

REVIEW



Interprofessional implementation of the Global Malnutrition Composite Score quality measure

Matthew L. Bechtold MD¹  | Kenneth G. Nepple MD² |
Sharon M. McCauley MS, RDN³ | Christina Badaracco MPH, RDN⁴ |
Ainsley Malone MS, RDN⁵ 

¹Division of Gastroenterology,
Department of Medicine, University of
Missouri, Columbia, Missouri, USA

²Department of Urology, University of
Iowa Hospitals and Clinics, Iowa City,
Iowa, USA

³Academy of Nutrition and Dietetics,
Chicago, Illinois, USA

⁴Evidence & Strategy, Avalere Health,
Washington, District of Columbia, USA

⁵Clinical Practice, The American Society
for Parenteral and Enteral Nutrition,
Silver Spring, Maryland, USA

Correspondence

Ainsley Malone, MS, RDN, 1356
Windtree Ct, New Albany, OH 43054,
USA.

Email: ainsleym@nutritioncare.org

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Abstract

Malnutrition in hospitalized patients can impact health outcomes, quality of life, and health equity. Quality improvement initiatives and quality measurement can help improve the care of those hospitalized patients with malnutrition. The new Global Malnutrition Composite Score (GMCS) was recently adopted by the Centers for Medicare & Medicaid Services (CMS) as a health equity-focused measure. Beginning in 2024, the GMCS is available for reporting through the CMS Hospital Inpatient Quality Reporting Program. The GMCS provides an opportunity to elevate the importance of patient nutrition status and evidence-based interventions throughout the interdisciplinary hospital decision-making process. To promote this opportunity, the American Society for Parenteral and Enteral Nutrition (ASPEN) held an “Interprofessional implementation of the Global Malnutrition Composite Score” webinar as part of its 2022 Malnutrition Awareness Week programming. This article summarizes the underlying rationale and significance of the GMCS measure and showcases clinical observations about integrating quality improvement and measurement into the acute care setting, as presented during the webinar.

KEYWORDS

acute care, Centers for Medicare & Medicaid Services, health equity, healthcare outcome assessment, hospitalization, malnutrition, quality improvement

INTRODUCTION

In the United States, the focus on malnutrition in hospitalized patients and its impact on their health outcomes, quality of life, and health equity is increasing among providers, researchers, and policymakers. A recent systematic review by the Agency for Healthcare

Research and Quality concluded both that evidence exists for associations between malnutrition and increased mortality and prolonged length of hospital stays and that hospital-based interventions for malnutrition can reduce mortality and improve quality of life among malnourished patients.¹ Similarly, the National Institutes of Health included developing

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effective nutrition interventions that “improve health outcomes and quality of life” in goals for a 2022 *Malnutrition in Clinical Settings: Research Gaps and Opportunities Workshop*.² More recently, in August 2022, the Centers for Medicare & Medicaid Services (CMS) adopted the Global Malnutrition Composite Score (GMCS) as a health equity–focused measure available for reporting through its Hospital Inpatient Quality Reporting (IQR) Program beginning in 2024.³

CMS adoption of the GMCS brings clinical identification and intervention for malnutrition to the forefront because the GMCS is the first nutrition-focused quality measure included in a CMS payment program. This new reporting opportunity will further elevate the importance of patient nutrition status and evidence-based interventions throughout the interdisciplinary hospital decision-making process. To promote this opportunity, the American Society for Parenteral and Enteral Nutrition (ASPEN) held an “Interprofessional implementation of the Global Malnutrition Composite Score” webinar as part of its 2022 Malnutrition Awareness Week programming. The recorded webinar is available open-access for viewing. Specific webinar objectives were identified as follows⁴:

- Define the composite score measure and its focus on components of the nutrition care process: screening, assessment, diagnosis documentation, and implementation of a nutrition care plan.
- Review how adopting the measure and using a quality improvement process can benefit patient care, providers, and hospital outcomes.
- Discuss how the interprofessional team can partner with administrators to adopt and report on the composite score measure to improve patient outcomes and support health equity.

This article summarizes the underlying rationale and significance of the GMCS measure and showcases clinical observations about integrating quality improvement and measurement into the acute care setting, as presented during the webinar.

INTERPROFESSIONAL DEVELOPMENT OF THE GMCS

Today's need for quality improvement in malnutrition care across acute care settings reinforces the importance of quality measurement. An estimated 20%–50% of hospital patients are at risk for malnutrition,⁵ but malnutrition is diagnosed in <9% of hospital patients.⁶ This gap leads to undiagnosed and untreated patients

who, as a result, are at greater risk for adverse health outcomes, including increased risk of complications such as infections, longer lengths of stay, and increased readmissions.^{7,8} Costs of care are also higher: average costs for patients diagnosed with malnutrition may be up to \$9,300 higher (\$22,200 vs \$12,900) than those of patients without the diagnosis.⁹

To improve clinical malnutrition care, the Academy of Nutrition and Dietetics (“the Academy”), Avalere Health (a healthcare consulting firm), and other stakeholders developed and launched the Malnutrition Quality Improvement Initiative (MQii) in 2013. MQii innovations of an interprofessional toolkit,¹⁰ set of malnutrition-focused quality measures,¹¹ and learning collaborative¹² have been instrumental in interprofessional implementation of practice improvements. Specifically, resources such as the toolkit help identify how to engage members of the interprofessional team to work together to implement a quality improvement project specific to malnutrition care; such action provides a necessary baseline to collect data and report on the GMCS measure. The innovations have also yielded many publications and presentations documenting the results of successful implementation of MQii tools.¹³ The MQii model can also be adapted for different patient populations and in other care settings¹⁴; for example, the MQii toolkit has been applied to pediatric malnutrition care.¹⁵ The GMCS measure was developed to be specific to the hospital, and it is too early to know how it may be adapted to other patient populations. Although these various resources were developed and tested in the United States, many aspects could be adapted and applied in other countries. Indeed, as part of the Academy's Strategic Plan, the MQii is at the heart of its key principle to “have a global impact in eliminating all forms of malnutrition.”¹⁶

From the start, the MQii has focused on the development of malnutrition quality measures that could be implemented nationwide to improve patient care and health outcomes. The leading organizations of the MQii initially developed, piloted, and validated four electronic clinical quality measures (eCQMs) specific to hospital care of adults aged 65 years and older. The measures (screening, assessment, care plan development, and diagnosis) are interdisciplinary and were developed through a process of identifying gaps in malnutrition care.¹⁷ This work demonstrated the value of malnutrition quality measures in supporting continuous quality improvement for hospital-based malnutrition care.¹³ The four measures were submitted to the National Quality Forum (NQF) for endorsement and to CMS for adoption and were included on CMS' 2016 list of measures under consideration.¹⁸ However, after both NQF and CMS recommended combining the four measures into a single

composite quality measure, the measure steward (the Academy) and developer (Avalere Health) subsequently developed and validated the GMCS¹⁷ as a composite eCQM (Figure 1).¹⁹

The GMCS was endorsed by NQF²⁰ in 2021 and accepted by CMS in 2022 for its Hospital IQR Program.³ The Hospital IQR Program encourages hospitals to improve care through reporting on both mandatory and voluntary quality-measure performance and is used to determine hospital Medicare payment rates. The GMCS is the first nutrition-focused eCQM and the first composite eCQM in any CMS payment program. Hospitals participating in the Hospital IQR Program can begin to voluntarily report on the GMCS beginning in January 2024. Additional information and resources about the GMCS are available through the Academy,²¹ the MQii,²² and ASPEN.¹⁹

THE GMCS' POTENTIAL TO ADVANCE HEALTH EQUITY

CMS commented when it released the final Inpatient Prospective Payment System rule in 2022 (which included updates in the Hospital IQR Program) that it was adopting three health equity-focused measures to “address health care disparities in hospital inpatient care and beyond.” CMS noted that as one of those three health equity measures, the GMCS would help hospitals screen for and identify unmet needs to more comprehensively address and monitor key contributors of poor health outcomes among their patients.²³ Indeed, it has

been identified that the GMCS will advance multiple health equity priorities identified in *The CMS Equity Plan for Improving Quality in Medicare*²⁴; similarly, the GMCS aligns with priorities in the more recent *CMS Framework for Health Equity 2022–2032*.²⁵ Also, implementation of the GMCS could highlight the importance of an interdisciplinary approach to advancing health equity and could help support hospitals in meeting the new Hospital Commitment to Health Equity measure adopted in the Hospital IQR Program³ and The Joint Commission's new requirements to reduce healthcare disparities.²⁶

The GMCS can help support health systems in their efforts related to health disparities in other ways, too. First, it can help to improve identification of and intervention for malnutrition within the hospital. Data from MQii Learning Collaborative members have documented disparities in the burden of malnutrition across different racial and ethnic groups as identified by registered dietitian nutritionists (RDNs) (Figure 2). These data have also shown higher readmission rates for non-Hispanic Black patients with malnutrition than for non-Hispanic White patients with malnutrition (Figure 3).²⁷ Awareness of such disparities is a necessary first step for hospitals to address health disparities.

A second way the GMCS can support addressing health disparities is through helping to identify and intervene in the related condition of food insecurity. Food insecurity is defined as the household-level economic and social condition of limited or uncertain access to adequate food²⁸ and is a risk factor for

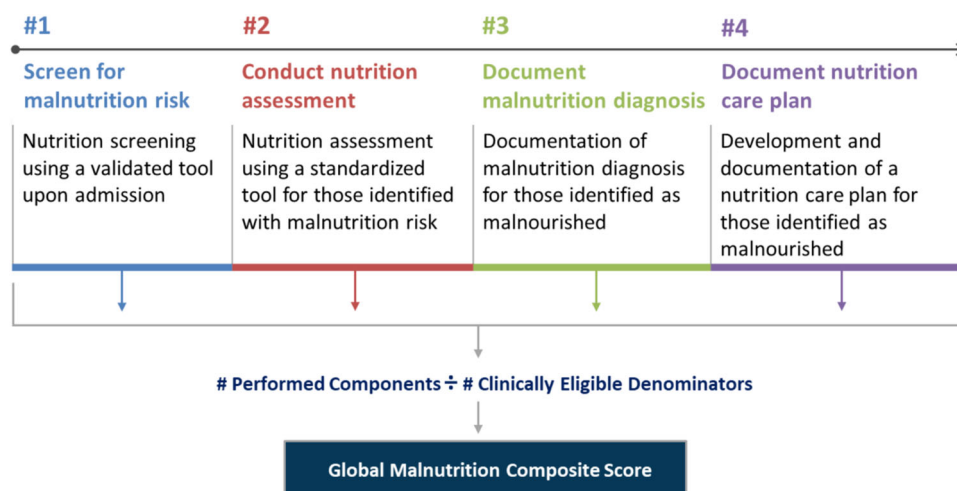


FIGURE 1 Elements of the nutrition care process reflected by individual components of the Global Malnutrition Composite Score. This figure reflects the measure specifications as of 2022. Adapted with permission from the MQii Global Malnutrition Composite Score Overview (2022).³⁹

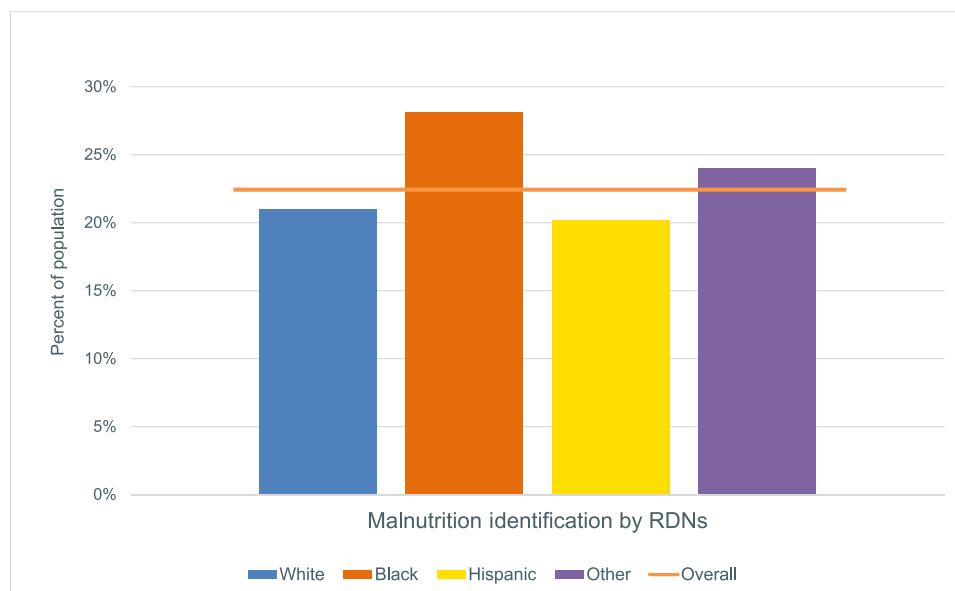


FIGURE 2 Malnutrition identification by RDNs in acute care hospitals, stratified by patient race and ethnicity, in 2019. RDN, registered dietitian nutritionist. Adapted with permission from the Commission on Dietetic Registration.²⁷

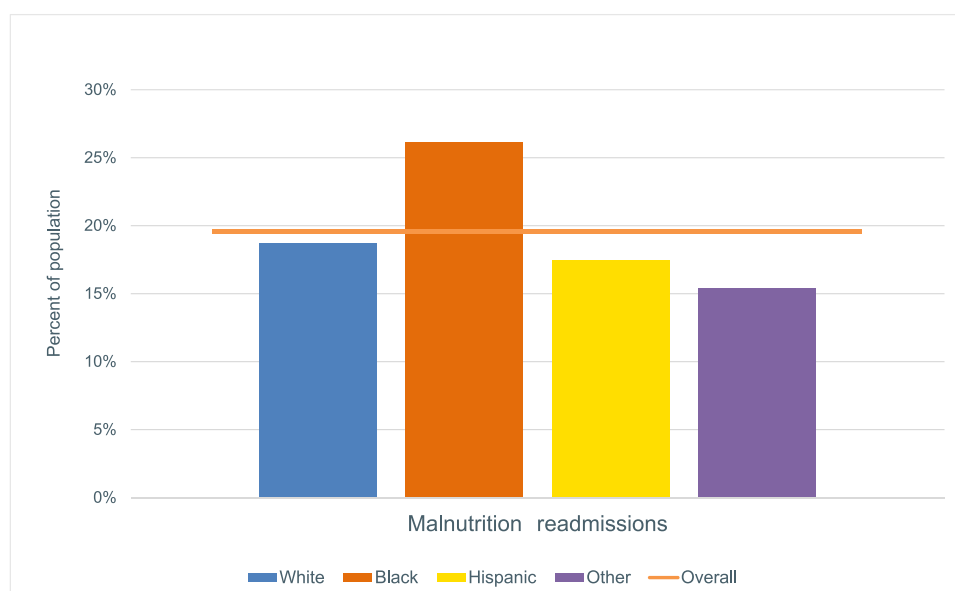


FIGURE 3 Hospital readmissions for patients with malnutrition, stratified by race and ethnicity, in 2019. Adapted with permission from the Commission on Dietetic Registration.²⁷

malnutrition.^{29,30} The GMCS aligns with the recommended care pathway for addressing both food insecurity and nutrition risk, as outlined in the *MQii Roundtable Proceedings: Advancing Health Equity Through Malnutrition Quality Measurement*, which also identifies multiple innovations by health systems, healthy plans, community organizations, and other partners to address food insecurity.³¹ Furthermore, a standardized care process for identifying malnutrition in acute care can set the foundation to explore the causes of food insecurity

in greater detail.²⁷ Indeed, anchoring the clinical workflow by the GMCS component measures can help address nutrition-related postdischarge needs (Figure 4),¹⁹ which is particularly important for connecting patients who have malnutrition and food insecurity to institution- and community-based resources.

Third, because nutrition care planning is linked to discharge planning, the GMCS can build on health equity efforts to improve posthospitalization health outcomes. Hospital patients with malnutrition are at increased

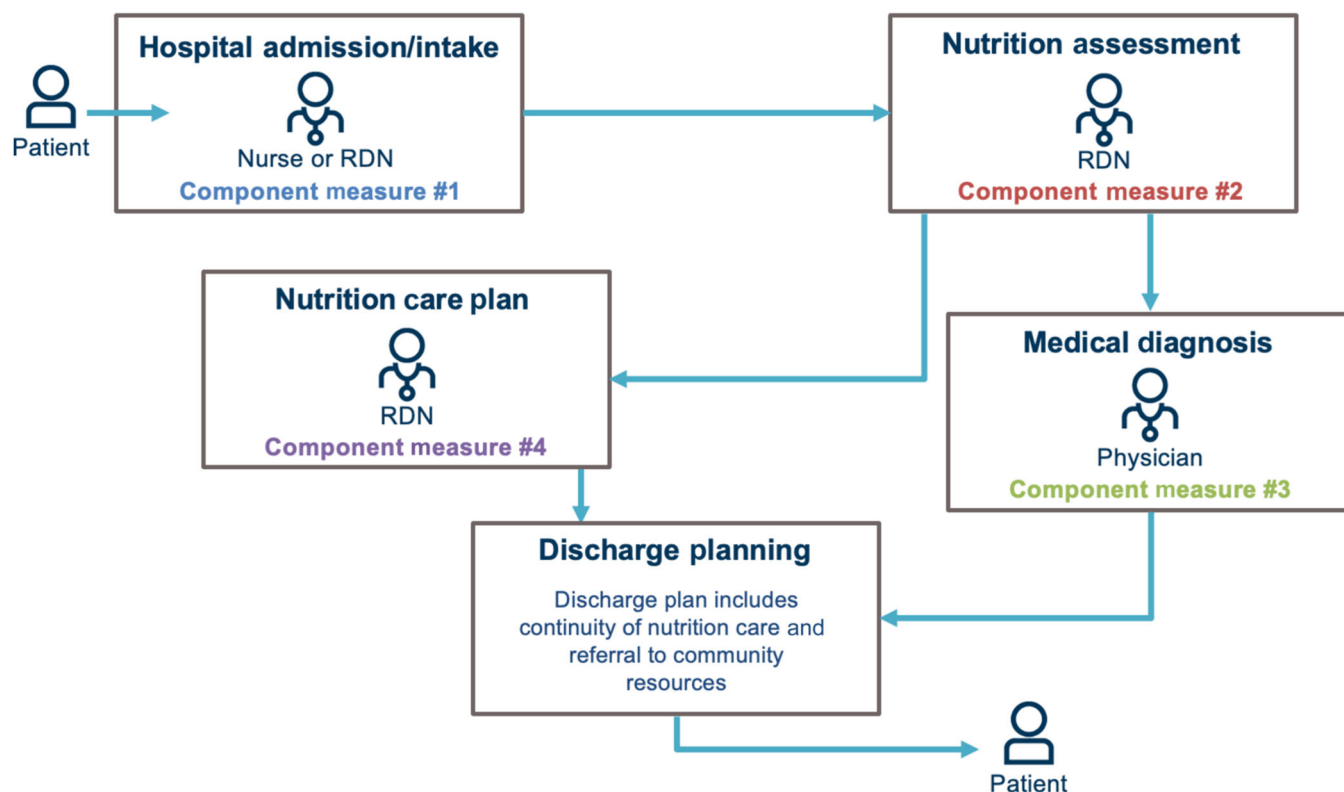


FIGURE 4 Anchoring clinical workflow by the Global Malnutrition Composite Score component measures can help address nutrition-related needs after discharge. This figure reflects the measure specifications as of 2022. RDN, registered dietitian nutritionist. Adapted with permission from the MQii Global Malnutrition Composite Score Overview (2022).³⁹

long-term risk for postdischarge mortality,³² emphasizing the importance of continued nutrition through subsequent settings of care. A recent meta-analysis and systematic review of randomized controlled trials suggested outpatient nutrition support after hospital discharge significantly increased nutrition intake and body weight and improved survival. The various forms of nutrition support included in the meta-analysis were dietary advice, food fortification, oral nutrition supplements, snacks between meals, enteral feeding, or a combination of these interventions.³³

THE BENEFICIAL RELATIONSHIP OF VALUE, MEASURE REPORTING, AND QUALITY IMPROVEMENT

CMS focuses on quality measurement across its programs to yield better healthcare value. In clinical practice, value is viewed as the relationship between quality and costs. To increase value, either quality increases or costs decrease. Poor healthcare quality can be reflected in costly outcomes such as increased lengths of stay and hospital-acquired conditions—both of which can be measured and tracked.

The evidence base for the value of effective nutrition interventions is growing. In 2017, ASPEN launched a value project to better understand the financial and quality impact of nutrition support therapy on high-priority therapeutic conditions. An initial study included a literature review (to identify evidence of clinical and economic impact of nutrition intervention on patient outcomes) and then analytic claims modeling in five therapeutic areas (sepsis, gastrointestinal cancer, hospital-acquired infections, surgical complications, and pancreatitis). The types of nutrition interventions considered in the study included direct provision of nutrition care (parenteral, enteral, and oral), indirect provision (diet modifications), nutrition care process (screening, assessment, and diagnosis), and nutrition education. The predicted annual cost savings of nutrition support therapy from the five therapeutic areas was \$580 million.³⁴

A subsequent value project study using similar methodology revealed nutrition support therapy has the potential to save at least \$104 million annually in treating patients who use Medicare and have hospital-acquired conditions.³⁵ A third value project study estimated that nutrition support therapy—specifically including widespread adoption of perioperative immunonutrition and

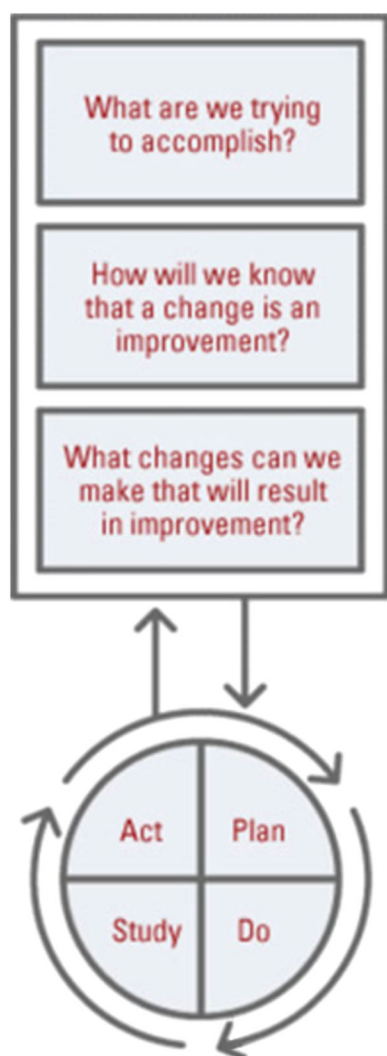


FIGURE 5 The Plan-Do-Study-Act model can be used to test implementation of process changes for quality improvement. Reprinted with permission from the Agency for Healthcare Research and Quality.⁴⁰

use of enhanced recovery pathways after surgery—in patients with gastrointestinal malignancies would save \$242 million annually.³⁶ Such evidence is important for informing clinical practice improvements as well as government rulemaking and payment structures.

One approach CMS uses to drive value-based healthcare is requiring that healthcare institutions report on adverse events and quality measure performance through programs such as the Hospital IQR Program. Institutions may then be financially compensated or penalized based in part on the number of specific adverse events they report and whether/how they perform on certain quality measures. Although it does take focused effort, this reporting has created incentives for institutions to improve clinical processes and health outcomes. Across the United States, hospitals and their clinicians

have responded by implementing quality improvement initiatives to change care processes. Ultimately, this reporting system supports CMS' three-part aim to achieve (1) better care for individuals, (2) better health for populations, and (3) lower costs.³⁷

Evaluation of quality improvement initiatives relies on measurable outcomes but differs from traditional clinical research in several ways: the aims of quality improvement studies specifically focus on improving care, sample sizes are often smaller, changes in approach/process are part of the study interventions and occur throughout sequential tests, and the results are specific to the teams engaged in the studies.³⁸ Detailed information and resources on implementing malnutrition quality improvement are available from the MQii.³⁹ Quality improvement can be guided by different models, including the Plan-Do-Study-Act (PDSA) model (Figure 5).⁴⁰

Using the four PDSA steps dissects the cyclical change process to allow for evaluation of the outcome(s), improvement on the care process, and then evaluation again.⁴⁰ Observations and learnings from implementing a clinical quality improvement study at the University of Missouri hospital are presented in Table 1.

IMPLEMENTING MALNUTRITION QUALITY MEASUREMENT

Malnutrition quality improvement programs yield better malnutrition care processes and health outcomes; they can also help prepare hospitals to collect data for tracking and reporting on a malnutrition quality measure. This was demonstrated through the University of Iowa hospital and clinics' experiences in piloting an improved malnutrition care workflow and implementing malnutrition quality measurement.

Malnutrition quality measurement can seem daunting, but importantly, it provides the framework for documenting incremental process improvement and validates the nutrition care process. Engaging all the relevant stakeholders in quality measurement is critical, beginning with the interprofessional team as well as advanced practice providers, electronic health record (EHR) vendors, and representatives from the informatics, finance, clinical documentation improvement, quality/safety, and patient discharge groups. Although every hospital is unique, the University of Iowa hospital and clinic stakeholders reported spending considerable time working with their informatics group to determine how different malnutrition-related data points aligned with clinician workflows, where those data resided in the EHR, who had access to the data, and what information

TABLE 1 Observations and learnings from implementing a clinical quality improvement study following the PDSA model at the University of Missouri hospital.

PDSA step	Observations and learnings
Plan	Identify the clinical process problem to be addressed and create an “aim” statement specifying the goal and how much improvement is anticipated within a given timeframe.
Do	Seek stakeholder engagement from the interprofessional team—including physicians, nurses, RDNs, and specialists.
Study	Collect background data/information to determine where there are roadblocks and what needs to be done to address the clinical process problem. Brainstorm with the interprofessional team and identify specific actions/interventions that could make a meaningful difference in the clinical process.
Act	Implement the specific actions/interventions and collect data on the outcomes—ie, did the process improvement action/intervention make a difference? Do not implement all specific actions/interventions at once (this can be too overwhelming and make it difficult to determine what may make a difference); instead, implement individual actions/interventions over time through successive PDSA cycles. Continue any action/intervention that makes a difference and keep it going. If an action/intervention does not make a difference, try another action/intervention until the process improvement changes are successful in achieving the specified goal.

Abbreviations: PDSA, Plan-Do-Study-Act; RDN, registered dietitian nutritionist.

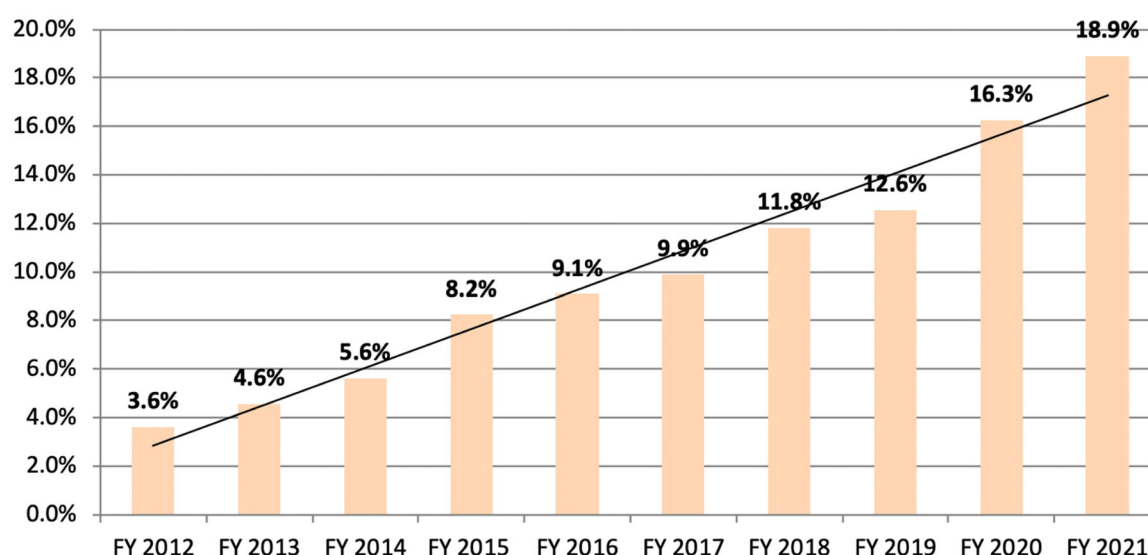


FIGURE 6 Continued improvement in annual inpatient malnutrition diagnosis rates at the University of Iowa hospitals and clinics. FY, fiscal year.

could be pulled from the data and shared with other team members to promote process improvement.¹²

Quality measure data should also be shared with hospital management to help build the case for adequate resources to deliver quality malnutrition care. The University of Iowa hospital and clinics conducted a pilot in 2016 to improve malnutrition care workflow and refine the process of inpatient evaluation, with a focus on communication and accurate malnutrition assessment in their EHR (using the Academy/ASPEN malnutrition

characteristics consensus statement).⁴¹ Before implementing the pilot, the institution estimated malnutrition was diagnosed in <10% of inpatients. The improved malnutrition care workflow was piloted in two inpatient units during a 4-month period. In the pilot, malnutrition was identified in 42% of patients in the two inpatient units, resulting in what was viewed as a favorable impact on allowable length of stay and related reimbursement. These positive results led to adoption of the improved malnutrition care workflow across the entire hospital

TABLE 2 Observations and learnings from implementing malnutrition quality-measure components at the University of Iowa hospitals and clinics.

Malnutrition quality measure	Clinician responsible	Observations and learnings		
		Advantages	Challenges	Solutions
Malnutrition screening	Nurse	• Nutrition screening questionnaire already included in EHR; completion required before nurse can advance to next EHR screen in initial nursing assessment • Based on nutrition screening results, EHR guidelines recommend patient referral to dietitian for nutrition assessment (within specified timeframe)	• Gaining access to nutrition screening data in EHR • Data accuracy (for example, EHRs for patients in ICU did not show initial nutrition screening, in part because most ICU patients routinely receive nutrition assessment)	• Worked with informatics to gain access to data • Worked with informatics to resolve accuracy challenge
Nutrition assessment	Dietitian	NA	• Nutrition assessments generally recorded as narrative notes vs electronic data and needed to be developed into technology-enabled workflow	• Worked with informatics to change nutrition assessment chart note format to discrete data, which were initially based on clinical characteristics in Academy of Nutrition and Dietetics/ASPEN malnutrition consensus statement and now also include GLIM criteria
Diagnosis	Physician	NA	• Communicating results of dietitian assessment to physician • Increasing physician understanding of importance of documenting malnutrition diagnoses and reviewing nutrition care plans	• Worked with informatics to develop EHR dot phrase, which can pull condensed version of nutrition assessment into physician progress notes and serve as prompt for physician documentation of malnutrition diagnoses • Continued physician education on importance of documenting malnutrition diagnoses so electronic data points can be captured
Nutrition care plan	Dietitian	• Increased visibility of malnutrition measures over time has garnered broader hospital-level support to develop an Advanced Nutritional Consult Service (staffed by gastroenterologist, dietitian, and pharmacist) that helps manage complex nutrition care, including transitioning patients to outpatient care	• Care plan available in EHR but not readily able to be abstracted electronically as discrete data points	• Worked with informatics to develop workflow template that differentiates among specific nutrition interventions recommended in nutrition care plan

Abbreviations: ASPEN, American Society for Parenteral and Enteral Nutrition; EHR, electronic health record; GLIM, Global Leadership Initiative on Malnutrition; ICU, intensive care unit; NA, not applicable.

and also hospital administration support for adding six full-time equivalent RDNs. With the improved malnutrition care workflow, the hospital's malnutrition diagnosis rate has continued to increase over time (Figure 6) and enabled the institution to better address patient malnutrition needs.

The University of Iowa hospital and clinics also participated in a study to evaluate the four malnutrition eCQMs initially developed through the MQii. The study assessed the feasibility, reliability, and validity of these measures and reported on specific process improvements that were implemented to educate staff and modify how information was documented in the hospital's EHR.¹² Further observations and learnings from their quality-measure implementation journey are reported in Table 2.

CONCLUSION

Interprofessional teams need to implement malnutrition quality improvement to maintain relevance in today's healthcare environment, which increasingly prioritizes value-based care. Effective quality improvement requires transitioning clinical nutrition workflows to discreet data and then using those data to support documentation of improved outcomes and quality-measure reporting. The GMCS measure provides further incentive for hospitals to either begin or build on their malnutrition quality improvement journeys. Finally, malnutrition quality improvement projects—starting even with small changes, when implemented consistently—can substantially benefit patient health and hospital outcomes.

AUTHOR CONTRIBUTIONS

Matthew L. Bechtold, Kenneth G. Nepple, Sharon M. McCauley, Christina Badaracco, and Ainsley Malone contributed to conception and content of this manuscript. All authors critically reviewed and revised the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

Matthew L. Bechtold is a speaker for Exact Sciences and the Nestle Nutrition Institute and on the Medtrition Advisory Board. Christina Badaracco is an employee of

Avalere Health. The remaining authors declare no conflict of interest.

ORCID

Matthew L. Bechtold  <http://orcid.org/0000-0002-0205-3400>

Ainsley Malone  <http://orcid.org/0000-0001-8345-0303>

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